

Sprint 6 – Missile Shoot with State and Collision

Student Information

Integrity Policy: All university integrity and class syllabus policies have been followed. I have neither given, nor received, nor have I tolerated others' use of unauthorized aid.

I understand and followed these policies: ☒ Yes ☐ No

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Date: 3/12/2024

Submission Details

Final **Changelist** number: 216155

Verified build: ☒ Yes ☐ No

Required Configurations: Debug only

YouTubeLink: <https://www.youtube.com/watch?v=d9AaViEnhNk>

Discussion (What did you learn):

Learned about states and setting up bumpers to switch states

Verify Builds

- Follow the Piazza procedure on submission
 - Verify your submission compiles and works at the changelist number.
- Verify that only MINIMUM files are submitted
 - No – Generated files
 - *.pdb, *.suo, *.sdf, *.user, *.obj, *.exe, *.log, *.pdb, *.db, *.user
 - Anything that is generated by the compiler should not be included
 - No – Generated directories
 - /Debug, /Release, /Log, /ipch, /.vs
- Typical files project files that are required
 - *.sln, *.csproj, *.cs,
 - App.config, AssemblyInfo.cs, CleanMe.bat
 - Resources Directory:
 - *.tga, *.dll, *.wav, *.gls, *.azul

Standard Rules

Submit multiple times to Perforce

- Submit your work as you go to perforce several times
 - Design Patterns – no minimum
 - Sprints – minimum of 5 real submissions (generally 10+)
 - As soon as you get something working, submit to perforce
 - Have reasonable check-in comments
 - Points will be deducted if minimum is not reached

Submission Report

- Fill out the submission Report
 - No report, no grade

Code and project needs to compile and run

- Make sure that your program compiles and runs
 - Warning level 4
 - NO Warnings or ERRORS
 - Your code should be squeaky clean.
 - Code needs to work “as-is”.
 - No modifications to files or deleting files necessary to compile or run.
 - All your code must compile from perforce with no modifications.
 - Otherwise it's a 0, no exceptions

Project needs to run to completion

- If it crashes for any reason...
 - It will not be graded and you get a 0

No Containers

- Containers (No automatic containers or arrays)
- Template or generic parameters
- No arrays
 - You need to do this the old fashion way - **YOU EARNED IT**

Leave Project Settings

- Do NOT change the project or warning level
 - Any changing of level or suppression of warnings is an integrity issue

Simple C#

- No .Net
- We are using the basics
 - Types:
 - Class, Structs, intrinsic types (int, float, bool, etc...)
 - NO arrays allowed!
 - Basics language features
 - Inheritance, methods, abstract, virtual, etc...

No Debug code or files disabled

- Make sure the program has only active code
 - If you added debug code or commented out code,
 - please return to code to active state or remove it

Adding files to this project

- Make sure you add the files in the appropriate sub-directories
- Make sure any new files are successfully integrated into the project
- Make sure your new files are submitted to Perforce

Due Dates

- See Piazza for due date and time
- Submit program perforce in your student directory assignment supplied.
- Fill out your this **Submission Report** and commit to perforce
 - **ONLY** use Adobe Reader to fill out form, all others will be rejected.
 - Fill out the form and discussion for full credit.

Goals

- Ship movement state – MoveRight, MoveLeft, MoveBoth
- Missile state – Flying, Ready
- Bumpers – for ship movement (right/left extreme)
- Collision between Missiles and Alien Grid
- Collision between missile vs top wall
- Collision between Ship vs Bumper
- Movement of delta time and delta_x changes proportional to number of aliens

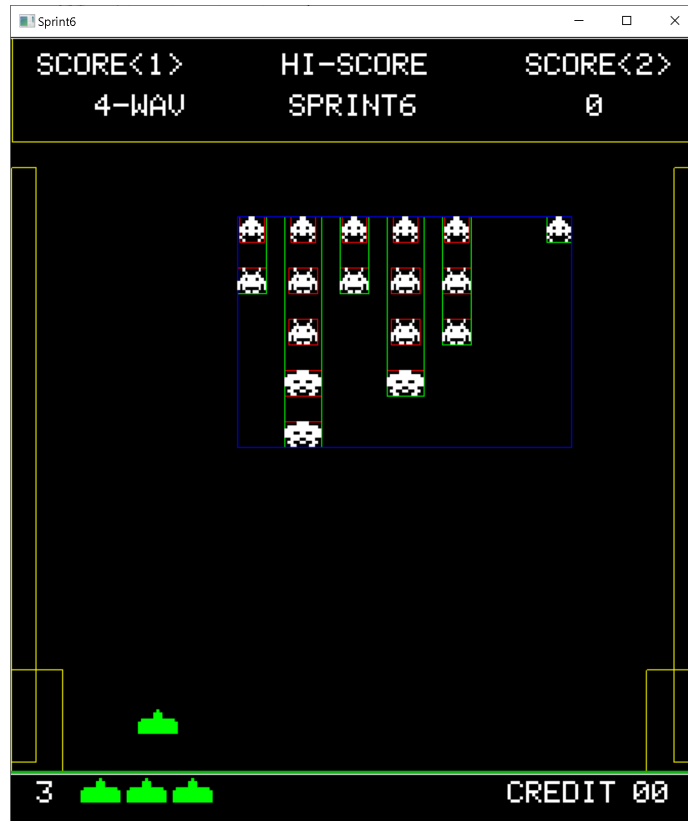
Assignments

Grading:

Summary of features: <https://youtu.be/q2yePm9t-7U>

- **Ship movement with Bumpers**
 - Ships are limited by the right/left bumper through collision and observer callback
- **Ship State**
 - Movement controlled by state
 - MoveRight, MoveLeft, MoveBoth
- **Missile State**
 - Can only shoot one missile at a time
 - States – Flying and Ready
 - Changes state from flying when the missile terminates through collision
 - Top Wall or Aliens
- **Missile vs Alien Grid**
 - Delete alien and missile to clear the grid
 - Deletes both the Game Object and corresponding Collision Object with collision box
 - No cheating by painting box Black
 - Changes Missile state from Flying to Ready
- **Missile vs Top Wall**
 - Deletes Missile and Changes missile state from Flying to Ready
- **Alien March Speed up**
 - Aliens start to move horizontally with 4 pixels per update
 - As the number of aliens goes from 55 to 0...
 - Horizontal Movement increase from 4 to 16 proportional to the number of aliens
 - Aliens start delta time (Drum Beat) is the time between movement and marching sounds
 - As the number of aliens goes from 55 to 0...
 - Delta time decreases from 0.7sec to 0.05sec portional to the number of aliens

Required Features:



Sizes and values need for this demo

- **Bumpers:**
 - 50x100 (w x h) on the right/left wall
 - Center at y=100 on the right/left side
- **Floor (bottom wall):**
 - 672x3 (w x h)
 - starting y=50
- **Ceiling (top wall):**
 - 672x103 (w x h)
- **Ship:**
 - 39x24(w x h)
 - (336,100) (x,y) start
 - **Speed of movement: delta_x = 2.0f**
- **Missile:**
 - 3x12 (w x h)
 - Starts at (ship.x,ship.y+20)
 - **Speed of movement: delta_y = 9.0f**
- **Reserve ships**
 - Reserve Ships (x,y) same as main ship 39x24(w x h)
 - (88, 33) (x,y)
 - (133, 33) (x,y)
 - (178, 33) (x,y)

- **Bottom Fonts**
 - Fonts are 24 pixels tall, and 15 wide.... exception "space" which is 3 pixels wide
 - (25, 30) (x,y) start // 3
 - (485, 30) (x,y) start // Credits
 - (608, 30) (x,y) start // 00
- **Grid Movement**
 - Movement proportional to number of aliens
 - As alien count decreases... the delta_time decreases and delta_x increases
 - Grid movement by delta_time
 - Start delta_time: 0.7s at 55 aliens
 - End delta_time: 0.05s at 0 aliens
 - Grid horizontal movement in x direction
 - Start delta_x: 4 pixels per update at 55 aliens
 - End delta_x: 16 pixels per update at 0 aliens

States - as described above affect the ship/missile

- Movement State
- Missile state

Demo – show in Video

- Needs to show states (movement and missile state)
- Need to show bumpers
- Need to show missile vs ceiling
- Need to show missile vs alien grid
- Need to show grid speed up proportional
- Need to show the bottom reserve ships and fonts

Approach

- Start with Sprint5 and modify and extend
- Reduce the problem to allow you to focus on a feature one at a time
- Mimic screen shot and video

Video Capture

- Have Visual Studio open and ready to go.
- Start recording
 - Show output window running
 - Speak clearly and loud
 - Discuss the features you added in this demo
 - Duration
 - Approx. 5-10 min long YouTube video capture
 - Do not exceed 10 min
 - Use Zoom recording
 - High resolution
 - Good sound quality

- Show on the capture... it compiling and then running
- Place YouTube link in PDF
- Make sure it public and runs without passwords or login
- Watch your video
 - Can you see the code clearly
 - Can you hear your voice clearly

Make sure you run CleanMe.bat before perform submission

Validation

Simple checklist to make sure that everything is submitted correctly

- Is the project compiling and running without any errors or warnings?
- Does the project runs **ALL** without crashing?
- Is the submission report filled in and submitted to perform?
- **Follow the verification process** for perform
 - Is all the code there and compiles “as-is”?
 - No extra files
- **Submit the YouTube link to PDF**

Hints

Most assignments will have hints in a section like this.

- Watch the lecture... look at the code supplied in class
- Study the projects in class lecture
- ASK questions... a lot of moving parts in this sprint

Troubleshooting

- Print to the output window to help you debug
 - It really helps
- Ask for clarification on piazza